No-Regret is Not No-Harm: Bandits with Consumable Action Sets

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Abstract

Sublinear regret is the standard justification for deploying sequential decision-making systems. We show that this justification is vacuous when actions consume the action set. We study bandits in which pulling an arm may destroy it, and in which each destroyed arm imposes a permanent negative externality on every subsequent reward. We prove that a policy achieving *exactly zero* regret against the best-available-arm benchmark can incur value loss $m\delta E$, unbounded in both the pool size m and the externality magnitude δE, and can drive total value negative while the optimal policy remains positive. The mechanism is that regret is measured against an optimum the learner has itself degraded. We further establish that the parameter governing optimal play, the expected marginal externality $\kappa_a = p_a e_a$, is (i) exactly the quantity that determines whether greedy is optimal, and (ii) subject to an estimation error floor $2\sigma/\sqrt{\min(T, \sum_j 1/p_j)}$ that is *independent of the horizon*, because each arm supplies exactly one before/after transition and the sample budget is capped by consumption rather than time. The quantity that matters most becomes least estimable exactly as it becomes more important. We give a closed-form correction, characterise an exactly-solvable special case, and release an open-source testbed.

1 Introduction

Consider an orchestrator routing tasks across API-backed tools. Some tools are robust; others trip a rate limit, exhaust a quota, or get a key revoked under heavy use, disappearing for the remainder of the session. Tools are not independent: those sharing upstream infrastructure—a vendor, an IP pool, a quota tier—degrade together when one is burned.

Table 1 shows two routers on identical instances. The first has a flawless record: at every step it pulls the highest-value available tool, so its regret against the best-available benchmark is identically zero. It ends on a fallback tool worth 0.470 per call with two tools remaining. The second reports regret of 0.25 per step—and is sitting at 0.782 per call, 66% higher, with four tools remaining.

The greedy router's zero-regret record is not evidence of good behaviour. It is an artefact of a benchmark that degrades in lockstep with the damage being done: every call greedy makes is optimal *given the ecosystem it has already ruined.*

Figure 1 shows the phenomenon in its cleanest form. As the coupling between arms strengthens, the greedy policy's cumulative regret stays pinned at exactly zero while its realised value falls monotonically and eventually turns negative—below the value of taking no action at all—even as the optimal policy remains comfortably positive. The two panels are computed on the same instances. A practitioner monitoring the left panel would see nothing wrong at any point.

This paper formalises that failure. Our contributions:

- **Four domain instantiations**—agent tool ecosystems, adaptive cancer therapy under competitive release, platform trials, and CMC design-space development—each simulated, with fit quality stated honestly (Section 9).
- **Theorem 2** (main result). A zero-regret policy can incur value loss $m\delta E$, unbounded in pool size and externality magnitude, and can finish net-negative while the optimum is positive. No-regret is not no-harm.
- **Theorem 1**. Greedy is optimal if and only if $\kappa_a = p_a e_a$ is constant across arms. Neither factor alone is the invariant.

Table 1: Two routers, identical seed and tools. The policy that looks worse by the standard metric achieves the better outcome.

router	step	realised	regret	tools left
greedy	0	1.150	0.0000	6
greedy	2	0.844	0.0000	4
greedy	7–13	0.470	**0.0000**	**2**
ECI	0	1.150	0.0000	6
ECI	1–4	0.988	0.0500	5
ECI	5–13	**0.782**	0.2500	**4**

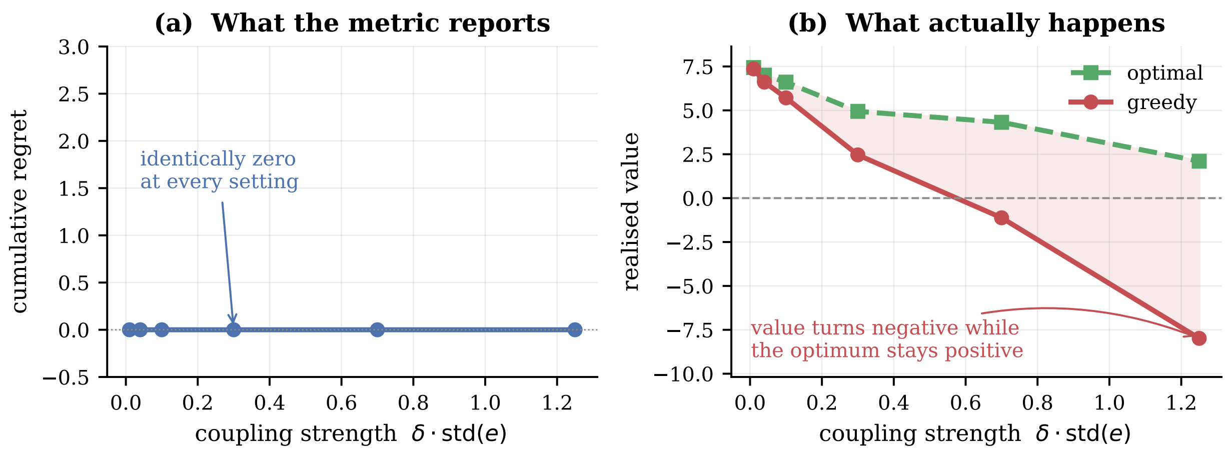

Figure 1: The central phenomenon. **(a)** Greedy's cumulative regret against the best-available-arm benchmark, computed exactly: identically zero at every coupling strength, because greedy never fails to pull the best arm currently available. **(b)** Realised value on the same instances. The shaded region is the loss. At strong coupling the zero-regret policy finishes net-negative while the optimum stays positive. Averages over 20 instances, $n = 6$, $T = 10$, exact dynamic programming.

- **Theorem 3**. The externality cannot be estimated to arbitrary precision: the error floor is horizon-independent, because the data-generating process is the consumption itself.
- **Theorem 5**. In the pure-sequencing regime the optimal order is ascending in e; arm values are irrelevant.
- A closed-form correction (ECI), a rollout policy reaching 99.5–99.9% of optimum, and an open-source testbed with one regression test per theorem.

2 Related work

Rotting bandits [4, 5, 6] model rewards that decay with the number of pulls (*rested*) or with elapsed time (*restless*); Seznec et al. [6] give a single adaptive-window policy near-optimal for both. Decay there is gradual rather than removal, and critically the learner is assumed to know which regime it is in. **Mortal bandits** [2] have arms with stochastic lifetimes, but expiry is exogenous. **Blocking bandits** make a pulled arm temporarily unavailable. **Sleeping bandits** [3] allow arbitrary exogenous availability. None of these couple arms: destroying one does not change another's reward distribution.

Bandits with knapsacks [1] is the closest structural neighbour: playing an arm consumes resources and the process halts when a budget is exhausted. The budget is global and exogenous, and consumption leaves other arms' rewards untouched. Here consumption permanently degrades *every* future reward. That difference is why the governing quantity is $\kappa = p \cdot e$ rather than a consumption rate.

Bandit learning with positive externalities [7] is the only prior work we are aware of that places externalities in a bandit model. There, satisfied users attract similar users, so externalities are positive and act on the *arrival process*; the authors show UCB incurs linear regret and develop balanced exploration. Our externalities are negative, act on *rewards*, and are triggered by irreversible destruction. The parallel is worth drawing: in both settings a standard algorithm fails badly, indicating that externality structure of either sign breaks classical guarantees.

Work on safe and conservative bandits constrains reward or maintains a baseline while leaving arms available; work on open multi-agent systems models exogenous population churn. Neither addresses action-triggered removal with reward coupling.

3 Setting

Arms $1, \ldots, n$. Arm a has mean reward v_a, destruction probability $p_a \in (0, 1]$, and externality coefficient $e_a \geq 0$. Let S_t be the set of available arms at round t and define the accumulated burden

$$B(S) \;=\; \delta \sum_{i \notin S} e_i.$$

Pulling $a \in S_t$ yields reward $v_a - B(S_t) + \eta_t$ with η_t sub-Gaussian with parameter σ, and destroys a with probability p_a, permanently adding δe_a to the burden. The horizon is T. The value function is

$$V(S, t) = \max_{a \in S} \Big[v_a - B(S) + p_a V(S \setminus \{a\}, t{+}1) + (1 - p_a) V(S, t{+}1) \Big].$$

Define the *expected marginal externality*

$$\boxed{\kappa_a \;=\; p_a \, e_a}$$

the expected permanent burden created by one pull of a.

Benchmark. We measure regret against the best available arm, $\max_{a \in S_t} \big(v_a - B(S_t)\big)$. This is the benchmark deployed systems use, and Theorem 2 is a statement about that practice. Value is measured against the exact optimum $V(S_0, 0)$, computed by dynamic programming over subsets for $n \leq 8$ and by rollout above that.

4 When is greedy optimal?

Theorem 1 (Characterisation). *Greedy—pull* $\arg\max_a v_a$*—is optimal if and only if* κ_a *is constant across arms.*

Proof sketch. Since $B(S)$ and $V(S, t{+}1)$ do not depend on the chosen arm, maximising the Bellman expression is equivalent to maximising $v_a - p_a D_a$ with $D_a = V(S, t{+}1) - V(S \setminus \{a\}, t{+}1)$. Decompose $D_a = \delta e_a L(S, t) + O_a(S, t)$ into a burden and an option component. The burden component contributes $p_a \delta e_a L = \delta \kappa_a L$, which is identical across arms exactly when κ is constant and hence drops out of the argmax.

Sufficiency. Under constant κ, couple the destruction randomness so a pair $\{a, b\}$ receives the same two draws in either order. The surviving-set distribution after both pulls is then identical, and since expected burden per pull is $\delta\kappa$ for both arms, the expected burden paths coincide. The only residual difference is which reward arrives first, giving $v_a - v_b \geq 0$. Iterating sorts any optimal policy into greedy order.

Necessity. The construction of Theorem 2 has $\kappa_H = E > 0 = \kappa_S$ and a strictly positive gap $m\delta E$. □

Two empirical checks. Varying p and e widely while pinning κ constant leaves greedy exactly optimal (0/10 instances broken, gap 0.000%); varying the product breaks it (10/10, gap 5.4–6.3%). Separately, the interchange inequality underlying sufficiency was audited exhaustively over all reachable states: 0 violations in 6,842 ordered pairs under constant κ, versus 56.2% violations when κ varies. The inequality holds precisely when the theorem says greedy is optimal.

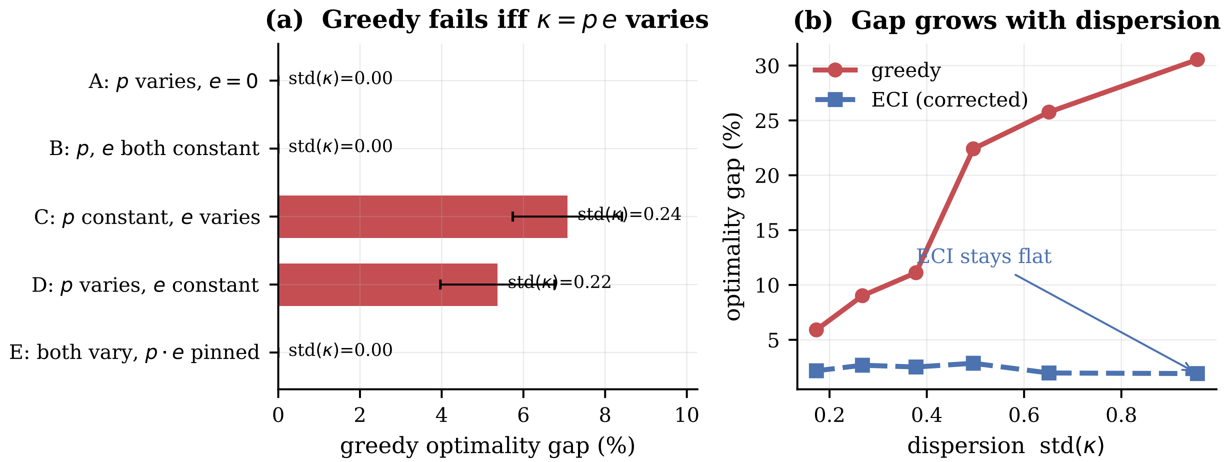

Figure 2: Theorem 1. **(a)** Five structural cases. Greedy is exactly optimal whenever $\kappa = p\,e$ is constant, and fails whenever it is not. Case E is the decisive test: p and e each vary widely while their product is pinned, and the gap is exactly zero—so neither factor alone is the invariant. Case D was a prediction of the refined conjecture and broke greedy as predicted. **(b)** The gap grows monotonically with the dispersion of κ, while the corrected index (Section 7.1) stays flat, so its advantage grows without bound. Error bars are standard errors over 12 instances.

The optimality gap grows monotonically with std(κ), from 5.9% to 29.1% across six bins over 240 instances. Figure 2 summarises both findings.

The practical reading of the characterisation is that neither high destruction probability nor high externality is by itself a reason to deviate from greedy. An arm that is fragile but harmless, or damaging but robust, can be treated normally. Only the *product*—the damage a pull is expected to cause—distinguishes arms for planning purposes.

5 A zero-regret policy can destroy unbounded value

Theorem 2 (Vacuity of no-regret under consumption)**.** *For any $M > 0$ there is an instance on which greedy attains regret exactly 0 against the best-available benchmark while $V^\star - V_{\text{greedy}} \geq M$. Moreover the loss can exceed $V^\star$, making the zero-regret policy net-negative while the optimum is positive.*

Proof. Fix $m \geq 1$, $E > 0$, $0 < \epsilon < \delta E$. Take $n = m+1$ arms: one *hot* arm H with $v_H = 1$, $e_H = E$, and m *safe* arms with $v_S = 1 - \epsilon$, $e_S = 0$. Set $p_a = 1$ for all arms and $T = n$, so each arm is pulled exactly once and only the order is chosen.

Greedy pulls H first since $1 > 1 - \epsilon$. H is destroyed, adding permanent burden δE, and each of the m subsequent pulls yields $(1 - \epsilon) - \delta E$:

$$V_{\text{greedy}} = 1 + m(1 - \epsilon - \delta E).$$

Deferring H to last yields $1 - \epsilon$ on each safe pull with no burden, and 1 on the final pull of H, whose burden is charged only to later rounds, of which there are none:

$$V^\star \geq m(1 - \epsilon) + 1.$$

Subtracting, $V^\star - V_{\text{greedy}} = m\,\delta E$, unbounded in m and in δE. Choosing $\delta E > 1$ gives $V_{\text{greedy}} < 0 < V^\star$.

At every round greedy pulls $\arg\max_{a \in S}(v_a - B(S))$, which is the definition of the benchmark; its instantaneous regret is 0 at every state. □

The closed form matches exact dynamic programming to machine precision at all 15 settings tested, $m \in \{1, 2, 4, 8, 12\}$ and $\delta E \in \{0.5, 1, 2\}$. At $m = 12$, $\delta E = 2$: $V^\star = 12.88$, $V_{\text{greedy}} = -11.12$, gap exactly 24.0000, regret 0.0000.

A correction, recorded. An earlier version of this construction claimed the gap grows linearly in T. That is false: with $p = 1$ the pool is exhausted after n pulls, so T beyond n is irrelevant. Under $p < 1$ the gap grows with T only until $T \approx \sum_j 1/p_j$, then saturates—the same sample-budget ceiling that appears independently in Theorem 3. The correct scaling is linear in pool size, which is the meaningful quantity: greedy's error is paid once per subsequent pull.

Interpretation. Greedy maximises v_a, the arm's *private* value. The optimal policy accounts for $\delta\kappa_a(T-t)$, the damage imposed on everything else. Regret is a private-value metric; under externalities private and social value diverge; therefore no-regret guarantees nothing about social outcome.

6 The externality cannot be learned

Theorem 3 (Estimation floor). *For any estimator of δe_i,*

$$\mathrm{SE}(\delta\hat{e}_i) \;\geq\; \frac{2\sigma}{\sqrt{\min\big(T,\ \sum_j 1/p_j\big)}},$$

which for $T > \sum_j 1/p_j$ is independent of the horizon.

Proof. All information about e_i enters through the level shift δe_i in the reward process at the moment arm i is destroyed. Conditioning on the destruction time with n_b rounds before and n_a after, the MLE is the difference of pre- and post-destruction sample means, with variance $\sigma^2(1/n_b + 1/n_a)$. By AM–HM, $1/n_b + 1/n_a \geq 4/(n_b + n_a) = 4/N$, with equality iff $n_b = n_a$, giving SE $\geq 2\sigma/\sqrt{N}$. Arm j survives Geometric(p_j) pulls, so $\mathbb{E}[N] = \sum_j 1/p_j$, capped at T. □

All three components verify numerically: the variance formula matches Monte Carlo to within 0.1%; the AM–HM floor is attained exactly at the balanced split (ratio 1.0000); and $\mathbb{E}[N] = \min(T, \sum_j 1/p_j)$ holds including when T binds.

The horizon-independence is stark, and Figure 3 makes it visible. At $p = 0.9$, increasing T from 20 to 320—a sixteenfold increase—changes RMSE by *zero* to four decimal places, because the pool caps the available data at 8.8 rounds no matter how long we wait. At $p = 0.1$ the round count climbs $20 \to 40 \to 71 \to 82.3 \to 83.2$, flattening exactly at the predicted $\sum_j 1/p_j = 80$, and the error curve flattens with it.

The consequence for practice is uncomfortable. In precisely the regimes where irreversible actions matter most—high p, where a single call can permanently remove a resource—the agent has the least opportunity to learn what those actions cost, because every observation is purchased by destroying the thing being priced.

Corollary 4 (The central tension). *As p rises, $\kappa = pe$ grows, so the externality matters more; simultaneously $N \approx n/p$ shrinks and SE $\geq 2\sigma\sqrt{p/n}$ degrades. The parameter governing optimal behaviour becomes less estimable exactly as it becomes more important.*

This is not a sample-complexity result that more data resolves. The data-generating process is the consumption itself: observing arm i's externality requires destroying arm i.

Structure does not rescue it. Suppose $e_a = \langle\psi, z_a\rangle$ for known features $z_a \in \mathbb{R}^k$, $k \ll n$, so each destruction informs a shared k-vector. Per-arm least squares is exactly worthless (identical to greedy to three decimals at every coupling level); feature-based estimation recovers only 17–28% of the available gap; and a *conservative* policy that assumes a uniform upper bound and learns nothing outperforms both at every level. If the mechanism were sample-sharing, capture should rise as k falls. It does not: capture is 4.5% at $k = 1$ and peaks at 30.3% at $k = 5$. At $k = 1$ the externality is nearly uniform, so κ has little dispersion and by Theorem 1 there is nothing to capture. The obstacle is dispersion, not sample size: the externality must vary for the correction to matter, and variation is what makes it hard to estimate.

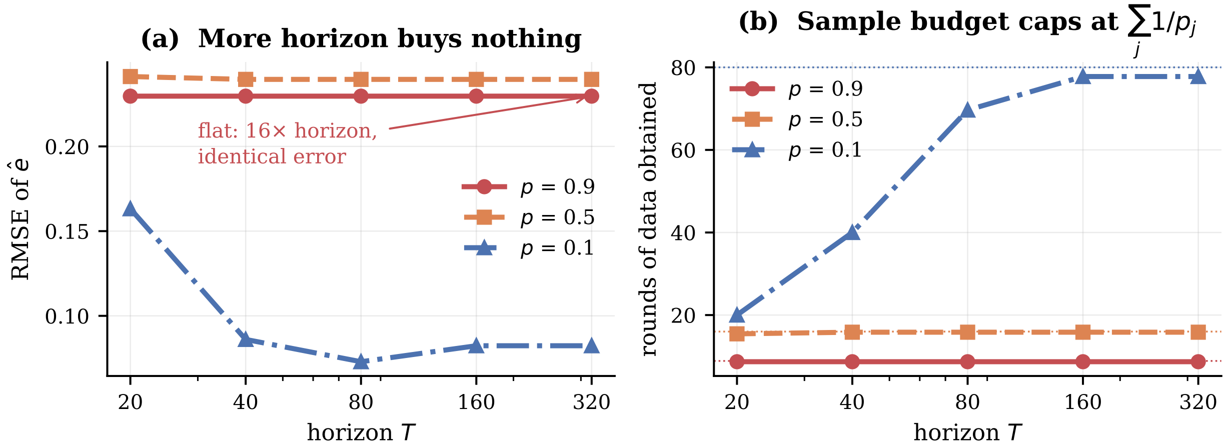

Figure 3: Theorem 3. **(a)** Estimation error for the externality vector as a function of horizon. At high consumption rates the curve is perfectly flat: waiting longer yields no additional information. Only at $p = 0.1$, where the horizon binds before the pool does, does more time help—and only until the pool becomes the binding constraint. **(b)** The mechanism. Dotted lines mark the predicted budget $\sum_j 1/p_j$; the realised number of rounds saturates there. The data-generating process is the consumption itself. Averages over 40 runs.

7 Corrections

7.1 The externality-corrected index

Theorem 1 hands over the correction. Pulling a creates expected permanent burden $\delta\kappa_a$ charged against every remaining round:

$$I(a,t) \;=\; v_a \;-\; \delta\,\kappa_a\,(T-t).$$

Because $B(S)$ is common to all arms it drops out of the ranking, so ECI is a true index: each arm's score depends only on its own parameters and global time, and the ranking is invariant to accumulated damage.

Against exact DP, ECI captures 52%–99% of the optimality gap, with capture improving as coupling strengthens. Its own gap is flat at 1.9–3.1% across the entire dispersion range while greedy's grows five-fold, so ECI's advantage grows without bound in $\mathrm{std}(\kappa)$.

7.2 Rollout

No closed form exists for the option-loss term ECI omits. One step of policy improvement over ECI captures it implicitly, since $V_{\pi_0}(S) - V_{\pi_0}(S \setminus \{a\})$ is exactly the marginal value of holding a. This reaches 99.5–99.9% of exact optimum, and with Monte Carlo policy evaluation runs at $n = 40$ where exact DP is infeasible, adding 3.3% over ECI at strong coupling.

7.3 An exactly solvable case

Theorem 5 (Pure sequencing)**.** *If $p_a = 1$ for all a and $T = n$, the optimal order is ascending in e, and arm values are irrelevant to the optimal order.*

Proof. Every arm is pulled exactly once, so $\sum_a v_a$ is a constant. Total value is $\sum_a v_a - \delta \sum_s e_{a_s}(n-s)$: an arm at position s pays its externality $(n-s)$ times. Minimising requires pairing large e with small $(n-s)$. $\square$

Verified by brute force over all $n!$ orderings, exact at 8/8 instances. Notably ECI is *not* exactly optimal even here—a conjecture we falsified—and the exact rule inverts the usual intuition: when consumption is certain, an arm's value is irrelevant to scheduling; only its externality matters.

Table 2: The regret and value columns are ordered backwards relative to each other.

policy	value	regret
greedy	14.449	**0.000**
Thompson sampling	15.171	1.341
UCB	13.257	6.136
conservative	17.089	8.160
ECI	**21.214**	10.069

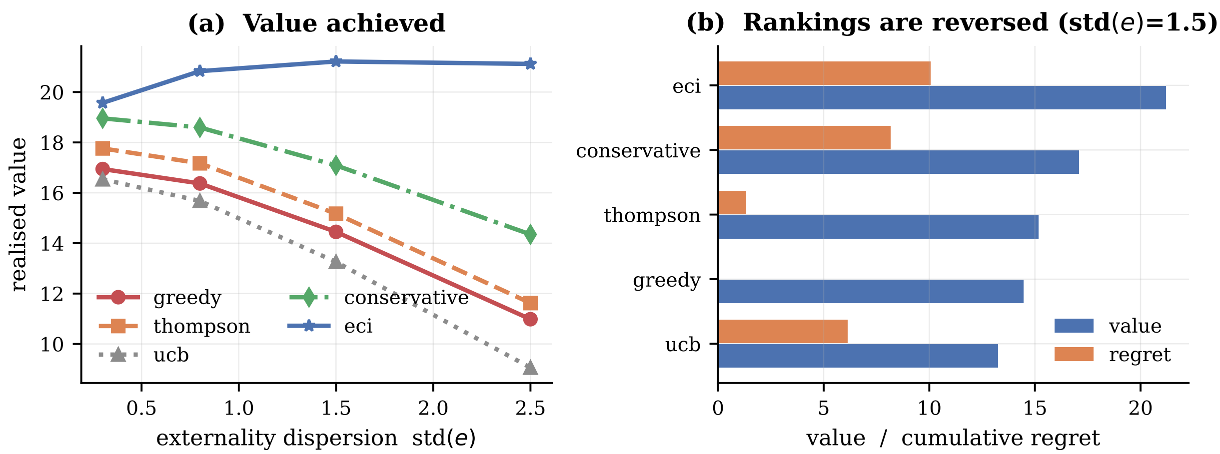

Figure 4: Policy comparison, $n = 20$, $T = 60$, 25 seeds. **(a)** Realised value against externality dispersion. The corrected index dominates throughout, and its margin widens as coupling strengthens. `conservative` uses no per-arm knowledge at all and still beats every consumption-blind method. **(b)** Value and cumulative regret side by side. The two orderings are close to reversed: `greedy` records the lowest regret and near-lowest value; `eci` records the highest regret and the highest value.

8 Experiments

Table 2 compares policies on 20 arms, 60 rounds, 25 seeds.

Figure 4 extends this across coupling strengths.

Thompson sampling is consistently *worse* than greedy under known parameters. Its exploration is not free here: sampling a suboptimal arm can destroy a low-value, high-externality arm. At strong coupling the corrected index nearly triples Thompson's realised value (20.8 vs. 7.2). The mechanisms compose rather than compete—TS handles uncertainty in v, ECI handles consumption—and TS+ECI dominates both.

9 Domains

The model is not specific to agents. We instantiate it in four settings, each supplying its own mapping to (v, p, e, δ) and each simulated rather than merely described. Table 3 gives the mappings, and we state the quality of fit honestly: two are close, one is a reasonable abstraction, and one is partial.

Adaptive therapy is the sharpest case. Under maximum tolerated dose, clinicians administer the highest tolerable dose, which is *exactly* the greedy policy. Higher dose intensity raises immediate tumour control v, but also raises both the chance of exhausting the drug-sensitive population (p) and the damage done when that happens (e), because the sensitive population was suppressing the resistant clone. Since v, p and e rise together, $\kappa = pe$ is highly dispersed—precisely the regime where Theorem 1 says greedy fails. The model therefore reproduces *competitive release* from first principles: it predicts that MTD is beaten by

Table 3: Domain instantiations. Each row is a running parameterisation, not an illustration.

domain	arm	destruction (p)	externality (e)	fit
Agent tools	API-backed tool	call trips a rate limit, exhausts a quota, or revokes a key	shared upstream infrastructure: burning one throttles the rest	strong
Adaptive therapy	dose level	dose exhausts the drug-sensitive cell population	released resistant clone degrades all later treatment	strong
Platform trial	treatment arm	arm dropped for futility on evidence the allocation generated	reliance on shared control; removal degrades concurrency for all comparisons	moderate
Design space	process parameter setting	run consumes scarce material or yields an out-of-spec batch	failure narrows the defensible filing envelope for all settings	partial

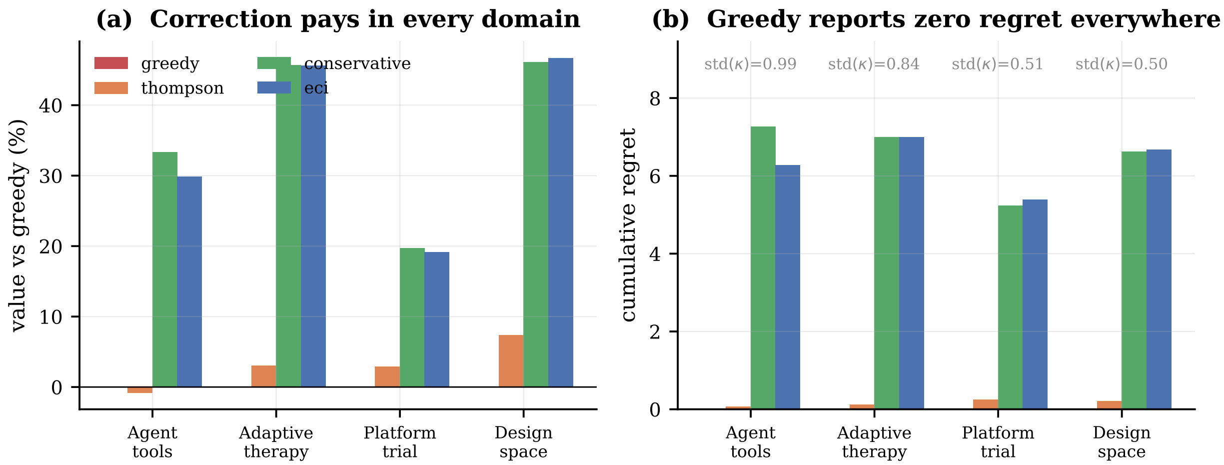

Figure 5: Four domains, 30 seeds each. **(a)** Value relative to greedy. Correction pays everywhere, most where κ dispersion is largest. **(b)** Cumulative regret. Greedy sits at exactly zero in all four domains while losing in all four. The consumption-blind baselines cluster near zero regret with it; the policies that actually perform are the ones the metric penalises.

a dose-modulating policy, which is what adaptive therapy trials find. We stress this is a reduced form; real tumour dynamics are a population process, not additive burden from dead arms.

Results. Figure 5 runs all four. Greedy records cumulative regret of exactly 0.0000 in every domain, and is beaten in every domain—by 30–49% where κ dispersion is largest. Thompson sampling, which is consumption-blind, gains at most 7% and in the agent domain is *worse* than greedy. The conservative policy, which uses no per-arm externality knowledge whatsoever, matches or beats the fully-informed index in three of the four domains—the practical form of Theorem 3.

9.1 Agent tool ecosystems in detail

Figure 6 returns to the routing example of Section 1 and shows the three quantities side by side over a single session. Panel (a) is what an operations dashboard displays; panels (b) and (c) are what is actually happening. The greedy router's regret trace is a flat line at zero for the entire session. Its value per call steps downward three times, and its tool count falls to one. The corrected router shows visible, non-zero regret throughout—and ends the session delivering more value per call with more tools still alive.

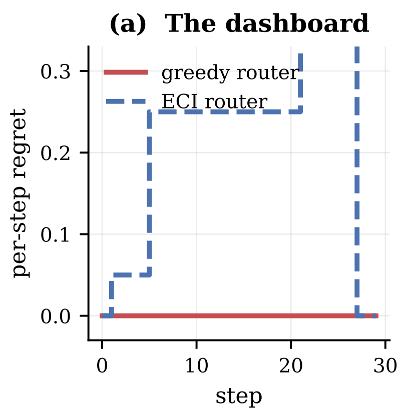

Figure 6: A single routing session, identical seed and tool set. **(a)** The metric: greedy's per-step regret is zero throughout, while the corrected router accrues visible regret. **(b)** Value per call: the router with worse regret delivers more. **(c)** Surviving tools: the router with worse regret preserves more of the ecosystem. The visible regret in (a) is the *price of preservation*, and the standard metric records it as a defect.

This is the practical form of Theorem 2. Nothing in panel (a) would trigger an alert, a rollback, or a model review. The failure is invisible to the instrument.

10 Robustness to the coupling form

The additive burden $B(S) = \delta \sum_{\text{dead}} e_i$ is a modelling choice, so we test it against four alternatives under exact dynamic programming: multiplicative (rewards scaled by $\prod(1-\delta e_i)$), saturating ($B_{\max}(1-e^{-\delta \sum e_i})$), networked (only graph-neighbours of the dead arm are harmed), and concave ($\delta(\sum e_i)^{0.6}$).

Theorem 2 is robust to all five. Greedy records regret of exactly 0.0000000000 under every coupling form while incurring positive value loss under every one (0.44–1.82). This follows from the proof: zero regret is a consequence of greedy's definition against the best-available benchmark, not of the burden's functional form.

Theorem 1 requires separability. Here the picture is more interesting. The characterisation holds exactly for additive, multiplicative and networked coupling (gap 0.000000% under constant κ), and fails slightly for saturating (0.135%) and concave (0.709%). Measuring the marginal burden of destroying one arm at different levels of accumulated damage identifies why: for additive and multiplicative coupling that marginal burden is invariant (0.15000 at zero, three and six units of prior damage), whereas for saturating and concave it falls as damage accumulates ($0.167 \to 0.107 \to 0.068$ and $0.150 \to 0.055 \to 0.043$ respectively).

Corollary 6 (Scope of Theorem 1)**.** *The characterisation holds exactly when the burden is* additively separable *across destroyed arms—when an arm's contribution does not depend on what else has already been destroyed. Under non-separable aggregation, constant κ no longer renders the burden term arm-independent.*

This is a sharper statement than the original, not a weaker one: it names the structural property the result requires. The degradation is also graceful—the residual gap under non-separable coupling is 0.14%–0.71%, against the 5%–12% gap greedy incurs when κ genuinely varies.

11 Limitations

Exact optimality results rest on dynamic programming at $n \leq 8$; larger-scale claims rest on rollout, and the coupling-robustness study of Section 10 inherits the same ceiling. Theorem 2 is a statement about the best-available-arm benchmark specifically—a forward-looking benchmark would not be fooled, but it is also not what deployed systems measure, which is the point. Theorem 1's sufficiency proof relies on a coupling argument whose formalisation under horizon truncation we verify exhaustively rather than prove in full generality.

The domain instantiations vary in fidelity. The agent and adaptive-therapy mappings are structurally faithful; the platform-trial mapping treats an inferential externality (loss of concurrent control) as if it were a reward penalty, and mixes patient benefit with information gain in a single scalar; the design-space mapping omits the terminal-commitment layer that defines the real problem—a one-shot declaration of design-space width that fixes the feasible set permanently. That layer is an optimal-stopping problem on top of the bandit and is, in our view, the most promising direction this work opens.

12 Conclusion

When actions consume the action set, regret is measured against an optimum the learner has already degraded. A policy can hold regret at exactly zero while driving value arbitrarily negative, and the parameter that would let it do better becomes unlearnable precisely as it becomes important, because acquiring each observation destroys the resource being priced. The practical consequence is direct: for irreversible actions, externality costs must be *specified*, not discovered. A crude uniform upper bound outperforms every learned estimator we tested.

Code. Library, all 24 experiments including the falsified ones, and one regression test per theorem: `https://github.com/<user>/consumable-bandits`

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